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Postexercise Dietary Protein Ingestion Increases Whole-Body Leucine Balance in a Dose-Dependent Manner in Healthy Children[1–3]

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Abstract

Background: Protein ingestion is important in enhancing whole-body protein balance in children. The effect of discrete bolus protein ingestion on acute postexercise recovery has yet to be determined.

Objective: This study determined the effect of increasing doses of ingested protein on postexercise whole-body leucine balance in healthy, active children.

Methods: Thirty-five children (26 boys, 9 girls; age range: 9–13 y; weight mean ± SD: 44.9 ± 10.6 kg) underwent a 5-d adaptation diet (0.95 g protein · $kg^{-1} \cdot d^{-1}$) before performing 20 min of cycling 3 times with a concurrent, primed, constant infusion of [^{13}C]leucine. After exercise, participants consumed an isoenergetic beverage (140 kcal) containing variable amounts of bovine skim-milk protein and carbohydrates (sucrose) (0, 5, 10, and 15 g protein made up with 35, 30, 25, and 20 g carbohydrates, respectively). Blood and breath samples were taken over the 3 h of recovery to determine non–steady state whole-body leucine oxidation (Leu_{OX}) and net leucine balance (Leu_{BAL}).

Results: Leu_{OX} (secondary outcome) peaked 60 min after beverage ingestion and demonstrated a relative dose-response over the 3 h of recovery (15 g = 10 > 5 > 0 g; $P < 0.001$). Leu_{BAL} (primary outcome) demonstrated a dose-response over the 3 h [15 g (24.2 ± 8.2 mg/kg) > 10 g (11.6 ± 4.3 mg/kg) > 5 g (5.7 ± 1.9 mg/kg) > 0 g (−3.0 ± 1.7 mg/kg); all $P < 0.01$] with all conditions different from zero (all $P < 0.001$).

Conclusions: Over the 3-h postexercise period, Leu_{BAL} was negative with carbohydrate ingestion alone; however, the co-ingestion of carbohydrates and 5 g high-quality dietary protein was sufficient to promote a positive postexercise whole-body protein balance in healthy, active children. Moreover, Leu_{BAL} increased in a dose-dependent manner within the protein range studied. Children should consider consuming a source of dietary protein after physical activity to enhance whole-body anabolism. This trial was registered at clinicaltrials.gov as NCT01598935. *J Nutr* 2017;147:807–15.

Keywords: leucine metabolism, dietary protein, exercise, recovery, muscle

Introduction

Two factors contributing to optimal growth and development during childhood are proper nutrition (1) and exercise (2). Physically active children have greater lean body mass than their sedentary peers (3). To support normal growth and this activity-induced augmentation of lean mass, adequate dietary protein ingestion is important for providing the requisite substrates to support the remodeling and growth of these lean tissues (including muscle) (1). Although it has been questioned whether current protein requirements are adequate for children (4) or whether recommendations may be influenced by habitual activity (5), research in adults has demonstrated that the amount of protein ingestion after exercise can affect muscle and whole-body tissue remodeling (6, 7). However, there is a relative dearth of information as to whether protein intake can similarly enhance the whole-body protein balance in active children and, if so, what the minimum intake to achieve this benefit may be.

We recently showed that postexercise protein ingestion induced a dose-dependent increase in whole-body protein balance over 9 h after an acute bout of exercise in healthy, active children (8). This relatively prolonged recovery (i.e., 9 h) was dictated by our

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[3] Supplemental Table 1 and Supplemental Figures 1–5 are available from the ''Online Supporting Material'' link in the online posting of the article and from the same link in the online table of contents at http://jn.nutrition.org.
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use of oral [^{15}N]glycine methodology and the time required to adequately collect the metabolic end products (9). As such, although our results demonstrated that protein ingestion after exercise can enhance the whole-body protein balance in a free-living state (8, 10), the inclusion of additional meals within the 9-h recovery period precluded our ability to determine the single-meal protein dose required to enhance protein balance after exercise in children. Therefore, given the potential for postexercise protein intake to enhance the whole-body protein balance in children (8, 10) and adults (11), we used an alternative stable-isotope methodology (i.e., [^{13}C]leucine infusion) that provides both greater sensitivity and time resolution to further our understanding of the efficacy of postexercise nutrition in enhancing anabolism in children.

Primed, constant infusions of [^{13}C]leucine and subsequent whole-body, non–steady state kinetics have been used extensively to measure protein metabolism after an acute meal ingestion at rest (12–14) and after exercise (11, 15) in adults. Despite the relative safety of this technique in children (16), it has been used only sparingly in pediatric populations in the fasted state and/or clinical settings (17–19) with no studies, to our knowledge, investigating discrete meal feeding. Alternatively, [^{13}C]leucine has been administered orally in rested children to determine the effect of small repeated-meal feeding on protein metabolism (20); however, this constant feeding approach generally does not reflect the discrete-meal protein ingestion that is characteristic of habitual feeding patterns and the associated non–steady state protein metabolic response (13, 21, 22). Finally, despite the extensive use of a [^{13}C]leucine infusion to model whole-body metabolism after exercise in adults (11, 15, 23), we are not aware of any studies that have applied this technique to address the postexercise nutritional requirements of active children. Therefore, to advance our understanding of pediatric-specific protein requirements to support protein remodeling after exercise, the primary objective of this study was to examine the effects of ingesting variable amounts of dairy protein on postexercise changes in whole-body net leucine balance (Leu_{BAL})[8] over a 3-h period. We hypothesized that, similar to studies in young adults (11, 15), protein ingestion after exercise would induce a positive Leu_{BAL} after exercise in children and that increasing protein doses would result in a graded dose-response in Leu_{BAL}.

Methods

Participants. Thirty-six healthy and recreationally active (as determined by medical and activity questionnaire) children (26 boys, 10 girls) volunteered to participate in the study. Before study completion, 1 participant (female) withdrew because of a refusal to eat the required controlled study diet and, therefore, did not partake in the experimental intervention visit. For inclusion in the study, subjects were required to be between the ages of 9 and 13 y, within 2 y from peak height velocity (24), have a minimum aerobic fitness of 35 mL · kg^{-1} · min^{-1} as assessed by the McMaster All-Out Progressive Continuous Cycling Protocol (see below), and be free of any existing medical condition (assessed via questionnaire). Exclusion from the study was made on the basis of the following criteria: use of medication at the time of the study, food allergy to milk proteins (e.g., whey or casein), and participation in another nutritional clinical trial within 2 mo of the current study. This study was approved by the Faculty of Health Sciences/Hamilton Health Sciences Research Ethics Board and was conducted in compliance with the standards set by the Declaration of Helsinki. Written, informed consent was obtained from each parent, and each subject provided written, informed assent before being enrolled in the study.

General overview. Enrollment in the study lasted for a total duration of ~3 wk, during which time subjects reported to the Child Health & Exercise Medicine laboratory at McMaster University on 3 separate occasions: a preliminary visit, a familiarization visit, and an exercise intervention visit. After the familiarization visit, subjects were randomly allocated (see below for details) to 1 of the 4 different parallel protein-based intervention groups (0, 5, 10, and 15 g). Consistent with previous studies in children utilizing [^{13}C]leucine tracers (20), participants were adapted for 5–8 d to a diet providing ~45 kcal energy · kg^{-1} · d^{-1} and 0.95 g protein · kg^{-1} · d^{-1} [the DRI for protein (25)]; this was done to minimize the adaptive metabolic loss of amino acids, which occurs primarily via an upregulation of oxidative pathways (26). The exercise intervention visits took place after this adaptation phase and were conducted in a randomized, double-blind, parallel, 4-group design in which each participant was assigned to 1 of the 4 allocated treatments groups. During the exercise intervention visit, subjects continued to consume a controlled diet similar to the adaptation phase and performed a bout of exercise in the morning. Within 15 min of exercise completion, subjects orally ingested one of the isoenergetic, milk-based protein beverages that contained a variable amount of protein (at a dose of 0, 5, 10, or 15 g) with additional carbohydrates. To assess our primary objective, changes in whole-body leucine kinetics were measured over 3 h by use of a primed, constant infusion of [1-^{13}C]leucine after the exercise bout and ingestion of the experimental beverage on the exercise intervention day.

Preliminary visit. A preliminary visit was performed, during which height (Harpenden wall-mounted Stadiometer), body mass (BWB-800; Tanita Corp.), percentages of body fat and lean body mass (InBody 520 bioelectrical impedance analyzer; BioSpace Co.), BMI (in kg/m^2), chronological age, and maturity offset were determined. Maturity offset was calculated as years from peak height velocity (24). Aerobic fitness was assessed by measuring peak oxygen uptake on a cycle ergometer (Lode Corival), as previously described (27). **Table 1** provides the physical characteristics of the subjects.

Familiarization visit. The purpose of the familiarization visit was to *1*) establish the individual resting metabolic rate; *2*) obtain blood samples for baseline metabolite enrichment (see below); *3*) allow the subjects to become familiar with the exercise protocol and equipment; *4*) verify appropriate exercise intensities; and *5*) confirm the exercise could successfully be completed by the subjects. At ~0730, subjects reported to the laboratory in an overnight-fasted condition. The individual resting metabolic rate was then assessed by using a calibrated metabolic cart (Vmax 29; SensorMedics) and an appropriately sized canopy while subjects rested quietly in a room for 10 min. Subjects were then provided with a light breakfast consisting of a small liquid meal (Boost Meal Replacement; Nestle Canada Inc.), as well as a minimum of 5 mL water/kg. Subjects then rested comfortably for 1 h before commencing the exercise protocol (using the same exercise protocol to be used during the exercise intervention visit). After the familiarization session, subjects were randomly allocated into protein-based intervention groups (0, 5, 10, or 15 g).

Exercise intervention visit. After an overnight fast, subjects reported to the laboratory at ~0730. A small breakfast consisting of ~24% and ~22% of total daily energy and protein intake, respectively, was consumed upon arrival. There were no differences between conditions for breakfast on the intervention day [either absolute (P = 0.45–0.99) or relative to body mass (P = 0.30–0.62)]. The macronutrient composition of the breakfast meal (mean ± SD) is provided in **Supplemental Table 1**. Thirty minutes before beginning the interval-based exercise protocol, a venous catheter was placed in the antecubital vein of one arm for blood sampling and was kept patent with periodic flushing with 2–3 mL saline. A second venous catheter was inserted in the contralateral arm for a priming dose of [1-^{13}C]leucine (7.6 μmol/kg, 99% atoms; Cambridge

[8] Abbreviations used: EAA, essential amino acid; Endo R_a, endogenous leucine rate of appearance; Exo R_a, exogenous leucine rate of appearance; Leu_{BAL}, net leucine balance; Leu_{OX}, leucine oxidation; Leu R_a, leucine rate of appearance; Leu R_d, leucine rate of disappearance; NB, net balance; NOLD, nonoxidative leucine disposal.

TABLE 1 Subject characteristics[1]

	Protein dose (g)			
	0	5	10	15
Sample size, *n* (boys)	10 (7)	8 (6)	9 (7)	8 (6)
Age, y	12.4 ± 1.2	12.4 ± 0.9	12.8 ± 1.0	12.1 ± 1.2
Height, cm	154 ± 11.5	157 ± 10.6	160 ± 9.7	156 ± 12.2
Height,[2] percentile	58.1 ± 28.9	68.8 ± 32.7	63.7 ± 35.5	70.3 ± 20.9
Body mass, kg	41.8 ± 9.6	44.2 ± 9.3	46.8 ± 9.8	48.2 ± 14.7
Body mass,[2] percentile	40.8 ± 24.1	51.8 ± 30.1	54.0 ± 33.3	63.4 ± 26.4
BMI, kg/m^2	17.4 ± 1.7	17.9 ± 2.5	18.2 ± 2.0	19.5 ± 3.6
BMI,[2] percentile	34.8 ± 20.6	45.0 ± 33.5	45.8 ± 28.7	57.6 ± 30.6
Lean body mass, kg	36.9 ± 8.5	38.0 ± 8.0	41.5 ± 10.2	39.7 ± 10.6
Body fat,[3] %	11.8 ± 4.7	14.2 ± 5.6	11.5 ± 5.3	16.6 ± 10.4
Maturity offset[4]	1.0 ± 1.1	1.0 ± 1.3	0.6 ± 1.2	1.1 ± 1.2
$\dot{V}O_{2peak}$, mL · kg^{-1} · min^{-1}	44.1 ± 5.3	44.7 ± 7.0	45.3 ± 7.2	40.5 ± 5.3
RMR,[5] kcal/d	1460 ± 288	1290 ± 183	1430 ± 185	1450 ± 259
RMR × activity, kcal/d	2180 ± 433	2090 ± 275	2150 ± 277	2180 ± 388

[1] Values are means ± SDs unless otherwise indicated. RMR, resting metabolic rate; $\dot{V}O_{2peak}$, peak oxygen uptake.
[2] Percentile rankings relative to 2000 CDC growth charts for the United States (28).
[3] Determined by using bioelectrical impedance analysis as described (29).
[4] Defined as the difference between the chronological age and the estimated age of peak height velocity, the latter of which was calculated as previously described (24).
[5] Assessed by accelerometry.

Isotopes) and 2.35 μmol $NaH^{13}CO_3$/kg before the initiation of the constant infusion of [1-^{13}C]leucine (7.6 μmol · kg^{-1} · h^{-1}). Subjects then began the interval-based exercise protocol, which consisted of 3 blocks of 20-min bouts of exercise with 5 min of rest separating the blocks. The 20-min exercise block consisted of cycling at 50% of their previously determined peak oxygen uptake with 10-s sprints at the workload corresponding to 100% peak oxygen uptake interspersed every 110 s between minutes 5 and 15 of exercise. Body mass (kg) was measured before exercise and during the rest period between exercise blocks. Subjects were provided with water to be consumed ad libitum to ensure adequate hydration (i.e., 100-mg body-weight change = 100 mL fluid).

After the interval-based exercise intervention, subjects rested comfortably for the remainder of the infusion. At 15 min postexercise, blood (10 mL) and breath samples were obtained before the ingestion of the randomly assigned experimental beverage. Each participant was provided a single 250-mL serving, which contained ether 0, 5, 10, or 15 g (depending on which of the 4 experimental groups he or she was assigned) of bovine skim-milk protein (containing both whey and casein protein fractions in a ratio ~1:4; Nestec, Ltd). Relative protein intake from the experimental beverage was 0, 0.12 ± 0.02, 0.22 ± 0.04, and 0.33 ± 0.08 g protein/kg body mass. Beverages were isoenergetic (~140 kcal) and provided a variable amount of carbohydrates (sucrose): 35, 30, 25, or 20 g, respectively. The experimental beverage was also enriched with [^{2}H$_3$]leucine to a concentration of 4% of the total leucine content of the beverage (assuming milk protein is ~10% leucine) to provide an estimate of exogenous leucine appearance from the dietary amino acids. Subjects were instructed to consume the beverage within 5 min of presentation. All participants tolerated the investigational beverages with no reported adverse events.

Subjects then rested comfortably in a seated position in the laboratory for 3 h, and additional breath and blood samples were collected at 30, 60, 90, 120, and 180 min after beverage ingestion (i.e., 45, 75, 105, 135, and 195 min postexercise) to determine fed-state leucine kinetics after a single meal, as described below. After the final sample collection, the infusion was terminated, and the catheters were removed.

Beverage preparation. All beverages were prepared on the evening before the trials by dissolving the powders provided by Nestec Ltd. (containing different ratios of carbohydrates and protein) into deionized water. Each beverage was reconstituted to a fixed volume of 250 mL. Beverages were kept chilled at 4°C until ingestion.

Controlled diet. Subjects were provided with a controlled diet providing 0.95 g protein · kg^{-1} · d^{-1} throughout the adaptation and intervention phases of the trial. There were no differences between conditions for diet during the 5-d adaptation phase [either absolute (P = 0.74–0.91) or relative to body mass (P = 0.11–0.66)]. See Supplemental Table 1 for total sample means ± SDs. The habitual activity levels of each subject (measured by accelerometer) were used to calculate energy intake, which varied throughout the trial to ensure that an approximately neutral energy balance was attained. All food was provided to the subjects at their home in prepackaged containers for each day of the study. Subjects and their guardians were given instructions for the subjects to not ingest any food or drink (except water) other than what was provided during the controlled diet period. To monitor dietary compliance, the subjects and guardians selected the food items that the subjects had consumed from a provided checklist and returned the containers to the laboratory to ensure consumption. Any food item uneaten during the controlled diet period (and subsequently returned) was recorded, and its nitrogen content was subtracted from the daily nitrogen intake.

Breath samples. For each breath sample, total carbon dioxide output was measured by indirect calorimetry over 10 min. Over this same sampling period, breath was collected directly from the breathing line into a 10-mL evacuated tube for determining the breath ratio $^{13}CO_2$to $^{12}CO_2$ by GC combustion isotope ratio mass spectrometry (BreathMat Plus; Finnigan MAT GmbH) by Metabolic Solutions Inc.

Blood samples. Blood samples were obtained from an antecubital vein via an indwelling catheter. At each blood collection time point, 10 mL whole blood was collected into a chilled EDTA-treated vacutainer and immediately placed on ice. Within 20 min of sampling, blood was centrifuged at 2000 × *g* for 20 min at 4°C to isolate the plasma before being stored at −80°C for future analysis. Plasma amino acid concentrations were determined by standard methods (30). Plasma glucose was analyzed by using a colorimetric assay (Cayman Chemical Company), and plasma insulin was measured by enzyme-linked immunoassay (Biosource; Medicorp). Plasma enrichments of [^{13}C]leucine and [^{2}H$_3$]leucine were measured by electron impact GC-MS (GC model 6890, MSD model 5973 network; Agilent Technologies) by monitoring ions *m/z* 327 and 328 for [^{13}C]leucine, *m/z* 232 and 233 for [^{13}C]ketoisocaproate, and *m/z* 282 and 285 for [^{2}H$_3$]leucine after conversion to their tert-butyldimethylsilyl derivatives. [^{13}C]ketoisocaproate enrichment was determined by GC-MS, as previously described (7). Plasma and breath enrichment data are summarized within **Supplemental Figure 1**.

As demonstrated by Boirie et al. (13), the determination of leucine oxidation (Leu_{Ox}) is not dependent on the existence of a steady state after ingesting a single meal. Therefore, non–steady state Leu_{Ox} was calculated from the equation:

$$Leu_{OX} = [(V_{CO2} \times Ei\,CO_2)/Ei\,KIC] \times [1/k] \quad (1)$$

where V_{CO2} represents the average CO_2 production (expressed as μmol · kg^{-1} · min^{-1}), Ei CO_2 represents the ^{13}C enrichment in expired CO_2 (expressed as μmol · kg^{-1} · min^{-1}), Ei KIC represents plasma α-[^{13}C] ketoisocaproate enrichment, and *k* represents the fractional bicarbonate retention factor chosen to be 0.80, as previously described (14, 21). Plasma α-[^{13}C]ketoisocaproate enrichment was chosen as the precursor because it is a good reflection of the immediate intracellular precursor for the irreversible decarboxylation of leucine (31).

The primary outcome of whole-body leucine balance (Leu_{BAL}) over 3 h after exercise and beverage consumption was then calculated from the equation (13):

$$Leu_{BAL} = \text{Leucine intake} - \text{AUC}\,Leu_{OX} \quad (2)$$

where leucine intake is the sum of [1-^{13}C]leucine intravenous infusion, dietary beverage leucine content (assuming 10% leucine content of milk protein), and oral [^{2}H$_3$]leucine tracer (provided at a concentration of 4% by weight of leucine ingestion). AUC Leu_{OX} represents the integrated AUC of Leu_{OX} over the entire 3 h postbeverage monitoring period, which represents the time period over which changes in leucine

metabolism would be most pronounced with discrete protein ingestion (13, 32) but would be similar to previous postexercise recovery durations in adults (7, 11). Leu_{BAL} over 3 h was converted to mg · kg^{-1} · min^{-1} by dividing by 180 to compare the effect size to previous studies in adults [(6, 11); see below].

Estimates of protein kinetics. To estimate changes in the rates of protein synthesis and breakdown in response to the ingestion of the protein-containing beverage (secondary outcomes), an oral tracer (i.e., [$^2H_3$]leucine at 4% of leucine ingestion) was consumed to estimate dietary leucine appearance for the calculation of non–steady state, whole-body leucine kinetics, as previously described (13). Briefly, measurements of the rates of leucine appearance (Leu R_a; μmol · kg^{-1} · min^{-1}) and disappearance (Leu R_d; μmol · kg^{-1} · min^{-1}) from the peripheral circulation, exogenous (dietary) (Exo R_a; μmol · kg^{-1} · min^{-1}) and endogenous (Endo R_a; μmol · kg^{-1} · min^{-1}) leucine appearance, and nonoxidative leucine disposal (NOLD; μmol · kg^{-1} · min^{-1}) were calculated by using the oral ([$^2H_3$]leucine) and intravenous ([1-^{13}C]leucine) labeled leucine tracers (13). Endo R_a was used as an estimate of whole-body protein breakdown, whereas NOLD was used to estimate whole-body protein synthesis. Net balance (NB) was then calculated as the difference between NOLD and Endo R_a (NB = NOLD − Endo R_a). Measurements taken immediately before beverage consumption (representing time 0 min) were calculated as static measurements, whereas all points after 0 min were determined by using non–steady state equations [developed by Proietto et al. (33)] described previously by Boirie et al. (13). To estimate the cumulative changes in measurements of whole-body leucine kinetics (e.g., Leu R_a, Leu R_d, Exo R_a, Endo R_a, Leu_{OX}, and NOLD), the AUC of each measurement over the entire 3-h period was calculated. Although we acknowledge that the most reliable means to assess the fate of dietary leucine is to use intrinsically labeled protein (13), this was not feasible for the present study nor a requirement for the primary outcome of Leu_{BAL}. This approach to providing free tracer within a carbohydrate and protein beverage to estimate Exo R_a is similar to that used in postexercise studies in adults (11).

Sample size. The estimated biological effect of consuming 5 g (estimated to be relative dose of ~0.13 g/kg) compared with 0 g protein was between Δ = 0.55 and Δ = 1.10 mg protein · kg^{-1} · min^{-1}, per previous studies in adults that used a 10-g (~0.13 g/kg) dose of protein (6, 11); therefore, the present study was powered to show a biological effect of Δ = 0.063 mg · kg^{-1} · min^{-1} in the whole-body leucine balance (assuming 9% leucine content of body protein) measured by [^{13}C]leucine infusion when subjects consumed the lowest protein intake. With the SD of the control group (0 g) estimated at 0.042 mg · kg^{-1} · min^{-1}, n = 8/group was determined to be required to show a difference between the nutritional strategy of 0 and 5 g of Δ = 0.063 mg leucine · kg^{-1} · min^{-1} as significant with a false-positive rate of 0.05, a 2-sided superiority test, and a power of 80%.

Randomization and blinding. Participant recruitment and testing occurred between November 2011 and March 2013. On enrollment in the study, the investigator in charge of beverage preparation (who had no other responsibilities within the trial) randomly assigned the subjects into 1 of the 4 experimental groups (**Figure 1**) using TrialSys, a Nestlé randomization program. Therefore, the person in charge of beverage preparation had knowledge of the subject randomization, the trial statistician configuring TrialSys was semiblinded, and the remainder of the study staff and participants were double-blinded. The randomization codes were broken only after all data were collected and cleaned and the statistical analysis plan was confirmed. The trial was completed when all participants were recruited and tested.

Statistics. For the primary outcome (Leu_{BAL}), a priori comparisons between 15 and 0 g, 10 and 0 g, and 5 and 0 g at 3 h after beverage consumption were conducted by using independent t tests. Exploratory analyses of the primary outcome (Leu_{BAL}) were performed with one-factor ANOVA (condition) to obtain comparisons between all beverage conditions that were not examined in the a priori analyses and, thus, further understand the dose-response relation. Differences between conditions for the controlled diet and the diet on the exercise intervention day were examined by using one-way ANOVA (condition). Changes

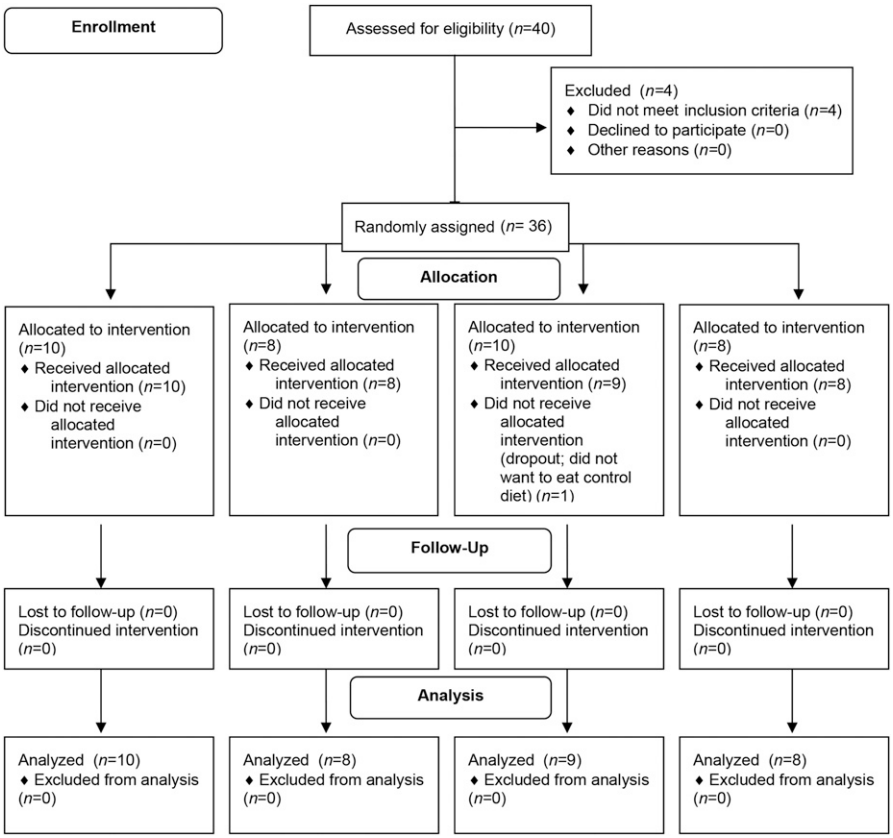

FIGURE 1 Consolidated Standards of Reporting Trials flow diagram. Flow of participants and study analysis.

in the AUC for the amino acid [e.g., leucine and essential amino acid (EAA)] concentrations and leucine kinetic parameters (e.g., Leu R_a, Leu R_d, Endo R_a, Exo R_a, Leu_{OX}, NOLD, and NB) were analyzed by using one-factor ANOVA (condition). Two-factor ANOVAs (condition × time) were used to measure time-dependent changes in plasma amino acid concentrations (e.g., leucine and EAA), whole-body leucine kinetics (e.g., Leu_{OX}, Leu R_a, Exo R_a, Endo R_a, Leu R_d, and NOLD), plasma insulin and glucose concentrations, and plasma (e.g., [^{13}C]leucine, [$^2H_3$] leucine, and [^{13}C]ketoisocaproate) and breath (e.g., [$^{13}CO_2$]) enrichments over the 3 h after exercise and ingestion of the experimental beverage. Differences between means for significant main effects or interactions were determined by using a Holm-Sidak post hoc test. Pearson's product-moment correlation coefficients were determined for Leu_{BAL} and leucine and protein intake over the 3 h. Pearson's product-moment correlation coefficients were also determined for intravenous [^{13}C]leucine enrichments and oral [$^2H_3$]leucine enrichments for each of the protein (i.e., 5, 10, and 15 g) conditions. Statistical analysis was performed by using SPSS Statistics version 20.0. Statistical significance was set at $P < 0.05$, and all data are expressed as means ± SDs unless otherwise indicated. No participant was excluded from the statistical test for each primary and secondary analysis per Figure 1.

Results

Whole-body leucine balance (primary outcome). Consistent with a priori comparisons, all protein conditions (i.e., 5, 10, and 15 g) differed from the 0-g condition ($P < 0.001$; **Table 2**). The difference between 0 and 5 g was ~76% of the planned effect [i.e., 0.048 compared with 0.063 mg leucine · kg^{-1} · min^{-1} (6, 11), respectively]. Exploratory analyses of the primary outcome by using one-factor ANOVA showed a main effect for condition for the absolute Leu_{BAL} over the 3 h ($P < 0.001$). All conditions were >0 g (−128 ± 93 mg; $P < 0.001$), with 15 g (1070 ± 146 mg) greater than both 10 g (514 ± 181 mg; $P < 0.001$) and 5 g (245 ± 61 mg; $P < 0.001$), and 10 > 5 g ($P < 0.001$). When expressed relative to body mass (**Figure 2**A), Leu_{BAL} (milligrams per kilogram) remained different between conditions ($P < 0.001$). When expressed relative to protein intake, the linear correlation (collapsed across all conditions) for the relative leucine balance had a line of best fit with the equation $y = 79.1\,x - 3.7$ and x-intercept = 0.05 g · kg^{-1} · 3 h^{-1} (Figure 2B). When expressed relative to leucine intake, the linear correlation (collapsed across all conditions) for the relative leucine balance had a line of best fit with the equation $y = 0.76\,x - 8.35$ and x-intercept = 10.96 mg · kg^{-1} · 3 h^{-1} (Figure 2C).

Leucine oxidation (secondary outcome). The AUC for Leu_{OX} demonstrated a main effect of condition ($P < 0.001$) (**Table 3**). Two-factor ANOVA revealed a main effect for condition ($P < 0.001$), time ($P < 0.001$), and the group × time interaction ($P < 0.001$) (**Figure 3**). Exploratory analysis of the relation between relative leucine intake and relative Leu_{OX} revealed that a segmental linear regression better explains the data than a simple linear regression ($P < 0.01$, $r^2 = 0.68$ compared with 0.58, respectively). The equation for the line of best fit below the breakpoint was $y = 7.2\,x - 0.31$, with the breakpoint occurring at x = 33.9 mg/kg (95% CI for the breakpoint: 24.9, 42.8 mg/kg). The slope above the breakpoint was equal to 0 (data not shown).

Plasma leucine and EAA concentrations (secondary outcome). The AUCs for plasma amino acid concentrations are presented in Table 3. The AUC for plasma leucine and EAA concentrations demonstrated a main effect for condition ($P < 0.001$). Time-dependent changes in plasma leucine and EAA concentrations are presented in **Figure 4**. Two-factor ANOVAs for both leucine and EAA concentrations revealed a main effect for condition ($P < 0.001$), time ($P < 0.001$), and the group × time interaction ($P < 0.001$).

Plasma insulin and glucose (secondary outcome). Time-dependent changes in plasma insulin and glucose concentrations are presented in **Supplemental Figure 2**. Data for total the AUC and peak plasma concentrations are collapsed and presented as group means ± SDs. The total AUC (4240 ± 1600 μU/mL; $P = 0.53$) and peak plasma insulin at 30 min (73.5 ± 29.9 μU/mL; $P = 0.97$) did not differ between conditions. The total AUC (13,300 ± 1260 mg/dL; $P = 0.22$) and peak plasma glucose at 30 min (91.1 ± 19.6 mg/dL) also did not differ between conditions.

Whole-body leucine kinetics (secondary outcome). The AUC of leucine kinetic data (Leu R_a, Endo R_a, Exo R_a, Leu R_d, and NOLD) are presented in Table 3. Briefly, all variables demonstrated a main effect for condition ($P < 0.001$ for all), except Endo R_a ($P = 0.75$).

Time-dependent changes in leucine kinetics are presented in **Supplemental Figures 3** and **4**. Total Leu R_a (Supplemental Figure 3A) demonstrated a main effect for condition ($P < 0.001$), time ($P < 0.001$), and the condition × time interaction ($P < 0.001$). All conditions were >0 g ($P < 0.001$); 15 g was >10 and 5 g ($P < 0.01$), with no difference between 10 and 5 g. Exo R_a (Supplemental Figure 3B) showed a main effect of condition ($P < 0.001$), time ($P < 0.001$), and the condition × time interaction ($P < 0.001$). All conditions were >0 g ($P < 0.001$); 15 g was >10 and 5 g ($P < 0.01$), and 10 g was >5 g ($P < 0.01$). Endo R_a (Supplemental Figure 3C) did not differ between conditions (main effect, $P = 0.28$) but showed a main effect of time ($P < 0.001$), and the condition × time interaction ($P < 0.01$). Total Leu R_d (Supplemental Figure 4A) showed a main effect of condition ($P < 0.001$), time ($P < 0.001$), and the group × time interaction ($P < 0.001$). NOLD (Supplemental Figure 4B) showed a main effect of condition ($P < 0.001$) and time ($P < 0.01$). There was no group × time interaction ($P = 0.25$).

Similar to Leu_{BAL}, analysis of NB with the use of one-factor ANOVA showed a main effect for condition ($P < 0.001$) (Table 3). NB was >0 for 15 and 10 g ($P < 0.05$ for both), tended to be >0 for 5 g ($P = 0.08$), and was <0 for 0 g ($P < 0.001$). The linear correlation ($r^2 = 0.73$) for NB across the 3 h expressed relative to beverage protein intake (collapsed across all conditions) was

TABLE 2 Leucine balance–planned comparisons between groups of children who consumed a postexercise isoenergetic beverage containing variable amounts of milk protein and carbohydrates[1]

	15 vs. 0 g	10 vs. 0 g	5 vs. 0 g
Leucine balance,[2] mg · kg^{-1} · min^{-1}	0.151 (0.126, 0.176)	0.081 (0.063, 0.098)	0.048 (0.038, 0.058)

[1] Values are means (95% CIs), n = 8–10. A priori planned comparisons of the difference in leucine balance between each protein dose (5, 10, and 15 g) and a protein-free control (0 g).

[2] Units for the primary outcome (leucine balance) reflect those from previous studies (6,11) and from which the present study was powered. All conditions were different from 0 g, $P < 0.001$.

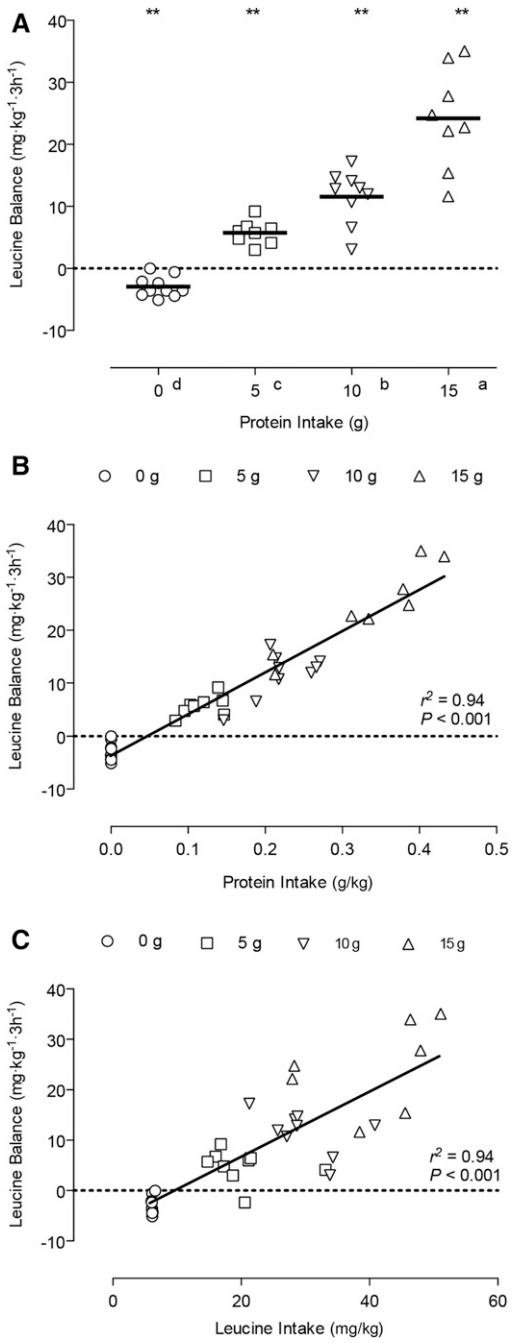

FIGURE 2 Leucine balance over the 3-h postexercise period in children who consumed an isoenergetic beverage containing variable amounts of milk protein and carbohydrates by group (A) and by individual response relative to total protein (B) and leucine (C) intake. In panel A, individual data points within a group are presented with the solid lines representing group means, n = 8–10. **Difference from zero balance, $P < 0.001$. Conditions with different letters are significantly different from each other; main effect for condition, $P < 0.001$. In panels B and C, the solid line represents the linear correlation (collapsed across conditions).

significant ($P < 0.001$). The equation for the line of best fit was $y = 373\,x - 45.5$, with x-intercept = 0.12 μmol/kg (data not shown).

Discussion

To our knowledge, we demonstrated for the first time in children that during the early (i.e., 3 h) recovery period after an acute bout of cycling exercise that Leu_{BAL} remains negative in the absence of postexercise protein ingestion, which is similar to previous studies in adults (11, 23). However, the ingestion of a single milk-protein and carbohydrate beverage containing 5 g of milk protein resulted in a positive Leu_{BAL} that increased in a dose-dependent manner within the protein range (i.e., 0–15 g) studied. This demonstrates that postexercise protein ingestion is required to induce a net anabolic environment after acute exercise in children.

Physical activity enhances the development of lean body mass in active children relative to their sedentary peers (3), which would necessitate a positive protein balance for net tissue growth. Consistent with adult studies (11, 23), we observed that the whole-body Leu_{BAL} was negative in the absence of protein ingestion. Although the apparently slightly less negative Leu_{BAL} in the present study compared with nonexercised children (18) and adults after similar aerobic exercise (11, 23) could suggest the exercise protocol provided an anabolic stimulus in our population, the negative balance in the protein-free control condition highlights the importance of adequate nutrition (i.e., protein ingestion) in enhancing net anabolism in active children. Our study clearly demonstrates that dietary protein ingestion is required to enhance Leu_{BAL} (and hence, protein balance) after exercise in children with as little as 5 g (~0.12 g/kg) protein inducing a positive anabolic environment during the early postexercise recovery period. Interestingly, the ingestion of 10 g (~0.13 g/kg) milk protein enhances but does not induce a net positive protein balance over 3 h after cycling exercise in adults (11). Similarly, based on reported Leu_{OX} rates, whole-body Leu_{BAL} after resistance exercise is not positive after the ingestion of 10 g (~0.12 g/kg) egg protein in adults (7). Collectively, this suggests that active, growing children have a relatively greater sensitivity to dietary amino acids during the early postexercise recovery period.

Consistent with the anabolic effects of dietary protein, our results further demonstrate that greater protein intakes (i.e., 10 and 15 g) induce a dose-dependent increase in net Leu_{BAL}. However, the linear increase in net Leu_{BAL} in the present study

TABLE 3 Amino acid concentrations and leucine kinetics in children who consumed an isoenergetic beverage containing variable amounts of milk protein and carbohydrates[1]

	Protein dose, g			
	0 (n = 10)	5 (n = 8)	10 (n = 9)	15 (n = 8)
Plasma leucine, mmol/L	16 ± 2[d]	23 ± 1[c]	30 ± 2[b]	36 ± 4[a]
EAA, mmol/L	122 ± 9[d]	153 ± 9[c]	193 ± 7[b]	212 ± 21[a]
Total Leu R_a, μmol/kg	359 ± 24[b]	476 ± 115[a]	500 ± 56[a]	545 ± 72[a]
Endo R_a,[2] μmol/kg	336 ± 24	368 ± 102	339 ± 59	344 ± 65
Exo R_a, μmol/kg	0 ± 0[c]	85 ± 20[b]	138 ± 31[b,a]	179 ± 83[a]
Total Leu R_d, μmol/kg	340 ± 23[b]	451 ± 106[a]	473 ± 53[a]	517 ± 69[a]
Leu_{OX}, μmol/kg	69 ± 12[b]	95 ± 17[b]	132 ± 28[a]	125 ± 22[a]
NOLD,[3] μmol/kg	290 ± 29[b]	380 ± 115[b]	367 ± 62[b]	419 ± 71[a]
NB,[4] μmol/kg	−45 ± 12[c]	12 ± 17[b]	28 ± 33[b,a]	76 ± 73[a]

[1] Values are means ± SDs. Data presented as AUC over 3 h after protein consumption. Conditions with different superscript letters are significantly different from each other; main effect for condition, $P < 0.05$. EAA, essential amino acid; Endo R_a, endogenous leucine rate of appearance; Exo R_a, exogenous leucine rate of appearance; Leu_{OX}, leucine oxidation; NB, net balance; NOLD, nonoxidative leucine disposal; Total Leu R_a, total leucine rate of appearance; Total Leu R_d, total leucine rate of disappearance.
[2] Estimate of protein breakdown.
[3] Estimate of protein synthesis.
[4] Calculated as the difference between NOLD and Endo R_a.

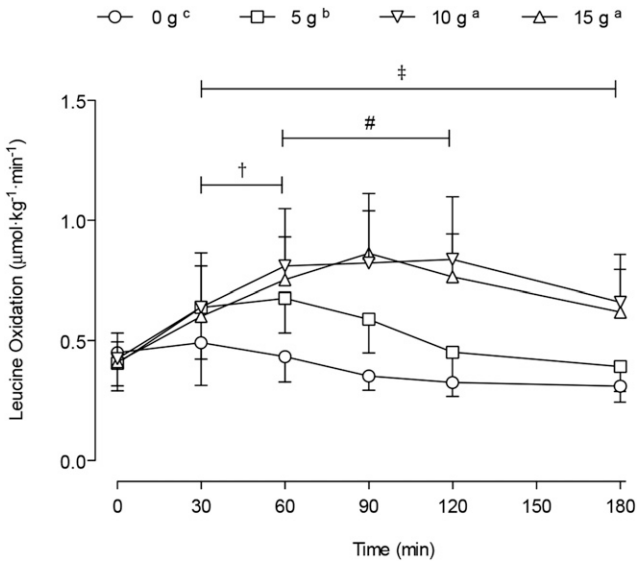

FIGURE 3 Total leucine oxidation over the 3-h postexercise period in children who consumed an isoenergetic beverage containing variable amounts of milk protein and carbohydrates. Values are means ± SDs, n = 8–10. Conditions with different letters are significantly different from each other; main effect for condition, $P < 0.001$. †,‡,#Difference from baseline values ($P < 0.05$): †5 g, ‡10 g, #15 g.

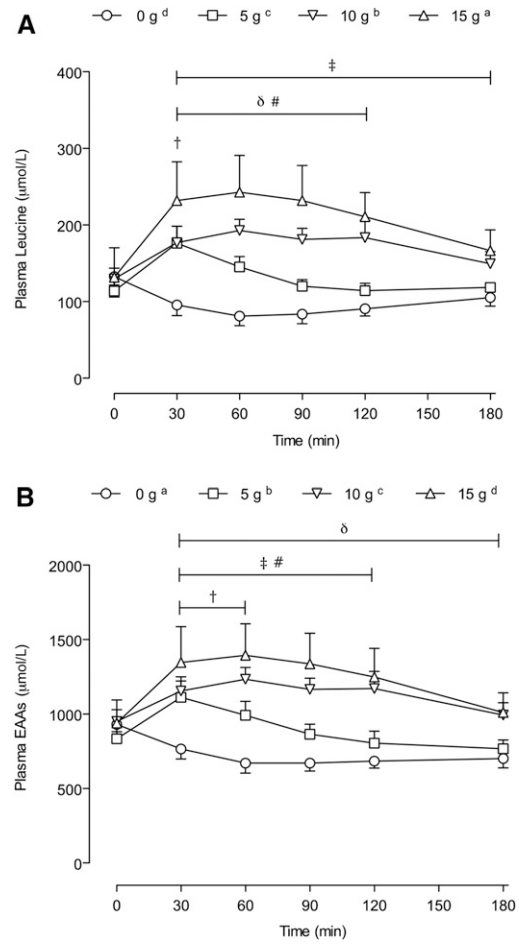

FIGURE 4 Leucine (A) and EAA (B) concentrations over the 3-h postexercise period in children who consumed an isoenergetic beverage containing variable amounts of milk protein and carbohydrates. Values are means ± SDs, n = 8–10. Conditions with different letters are significantly different from each other; main effect for condition, $P < 0.001$. δ,†,‡,#Difference from baseline values ($P < 0.05$): δ0 g, †5 g, ‡10 g, #15 g. EAA, essential amino acid.

may be influenced by the 3-h postprandial measurement period, which was dictated by the practical limitations associated with prolonged stable isotope infusions in children. For example, plasma leucine concentrations generally remained elevated at 3 h after ingestion of 10 and 15 g protein, suggesting the metabolic fate of these circulating free amino acids had not been defined by the end of our measurement period. Moreover, Leu_{OX} in adults has been reported to require ≤7 h to return to baseline after the ingestion of 30 g (~0.4 g/kg) isolated whey or micellar casein protein (13), suggesting utilization of a longer postexercise recovery period in the present study may have permitted a more accurate determination of Leu_{BAL} at higher protein intakes (i.e., >10 g). The plateau in total Leu_{OX} at the higher protein intakes could also suggest that a maximal oxidative disposal of this amino acid had been reached. Thus, the inability to capture the complete oxidative leucine disposal in the present study could have led to an underestimation of leucine lost from the body and, combined with the presence of a plateau in Leu_{OX}, may have resulted in an overestimate of Leu_{BAL} that is amplified at higher protein intakes. Given that the present study is the first, to our knowledge, to measure Leu_{OX} after bolus protein ingestion (either at rest or after exercise) in children, additional studies with longer postprandial periods or greater protein intakes or both would help clarify the physiological basis for the metabolic trends in Leu_{OX} and Leu_{BAL} observed herein. Therefore, although dose-dependent increases in Leu_{BAL} were observed in the present study, absolute protein intakes of ≤10 g might efficiently enhance whole-body protein balance yet minimize whole-body amino acid oxidation in our pediatric population.

Postexercise protein ingestion elicited an expected dose-dependent increase in plasma leucine and EAA concentrations with no difference in plasma insulin, the latter of which is likely related to the carbohydrate content of the beverages. Although it has been shown in rested children that insulin can enhance Leu_{BAL} in the fasted state through a mild suppression of Leu R_a (i.e., protein breakdown) (18), it is well established that concurrent protein ingestion is required to induce a net positive protein balance in adults at rest (13, 34) and after exercise (11). Because leucine is preferentially metabolized within lean tissue (including skeletal muscle) (35) and whole-body leucine kinetics qualitatively mirror that within skeletal muscle during the early postexercise recovery period in adults (11), it could be argued that the changes in leucine metabolism in the present study reflect, to some degree, the metabolism within skeletal muscle. In the present study, the Leu_{BAL} dose-response during the acute 3-h recovery period is in general agreement with studies in adults that demonstrate protein ingestion induces a dose-dependent stimulation of postexercise muscle protein synthesis (7, 36) and subsequently induces a net positive muscle protein balance (11). Therefore, although we are unable to determine the tissue(s) that contributed to the greater postexercise net Leu_{BAL} in the present study, we suggest that the dose-dependent increase in plasma amino acid concentrations was likely the major factor responsible for the observed increase in whole-body Leu_{BAL} after exercise in children.

The protein beverages in the present study were enriched with free [$^2H_3$]leucine to estimate the rate of appearance of dietary leucine to determine whole-body, non–steady state leucine kinetics and potentially provide insight into the underlying physiology responsible for changes in Leu_{BAL} (13). This approach

was taken given the cost and technical expertise required to generate intrinsically labeled dietary proteins (37–39), which are the acknowledged gold-standard tool to accurately determine digestion and absorption kinetics of whole proteins (13, 38). Although the appearance of free leucine in circulation is more rapid than dietary protein-derived amino acids during the immediate 30–60 min after a bolus ingestion of isolated protein (13), we hypothesized that the addition of carbohydrates to the beverages in the present study (and increased energy density) would slow gastric emptying (40) and ultimately minimize the discordant appearance rates between the added tracer and unlabeled dietary leucine. In partial agreement, we observed linear correlations between the change in plasma [$^2H_3$]leucine and [^{13}C]leucine enrichments in the 5- and 10-g conditions (**Supplemental Figure 5**), which would suggest our oral tracer generally modeled the fate of the unlabeled dietary amino acids (13); this was not the case with the 15-g protein intake, which leads us to interpret the changes in whole-body leucine kinetics in that group with caution. For example, the greater whole-body protein synthesis (i.e., NOLD) in 15 g compared with 0 g is likely a reflection of the more rapid appearance of the free tracer relative to the unlabeled dietary protein-derived amino acids, which would overestimate NOLD early in recovery (13) and inflate the 3-h aggregate response. Nevertheless, there was no effect of 5 or 10 g protein ingestion on NOLD, which is generally consistent with previous results in adults after exercise (11) and the primarily slowly digested casein protein in our milk-protein beverages (13, 32, 34). Alternatively, Endo R_a (i.e., a marker of body protein breakdown) was also not different between conditions in the present study, which we interpret as the result of a similar insulin response that would subsequently result in a consistent anti-proteolytic effect across groups (32). Despite the lack of robust effects on markers of whole-body protein synthesis and breakdown, the kinetic-derived estimation of Leu_{BAL} (i.e., NOLD − Endo R_a) revealed a dose-response qualitatively similar to our whole-body Leu_{BAL} (i.e., Leucine intake − Leu_{OX}). Collectively, these results could suggest that because of the rapid turnover rate of body proteins in active, growing children (41) only minor alterations in either protein synthesis or breakdown or both may be needed to translate into detectable difference in whole-body protein balance, as suggested previously (8, 10).

In conclusion, the addition of protein to a postexercise carbohydrate beverage was necessary to attain a positive whole-body Leu_{BAL} and protein balance during the early (i.e., 3 h) recovery period in children. Moreover, the postexercise consumption of high-quality protein increased whole-body Leu_{BAL} and protein balance in a dose-dependent manner. Therefore, similar to adult studies, children should consider consuming a source of dietary protein as part of a complete diet after physical activity to enhance whole-body anabolism. Future studies should delineate what effect the timing of protein ingestion relative to the exercise bout has on markers of whole-body protein balance and whether this early acute increase in NB would ultimately be additive to the 24-h response, the latter of which would be a requisite for the ability of postexercise protein ingestion to enhance overall lean tissue growth in active children.

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